Instruction Tuning

AI Engineering From scratch #10

What SFT actually does ?

- Supervised Fine-Tuning continues the same training loop from pre-training – forward pass, compute loss, backward pass, update weights – but on a different kind of data:
    - Instead of raw text, you train on structured conversations:
        - {“system”: “You are a helpful assistant.”, “user”: “What is the capital of France?”, …}

→ Pre-training gives the model knowledge. SFT gives the model manners.

Data formats

Three formats dominate the industry:

- Alpaca Format: { “instruction”: …, “input”: …, “output”: … }
- ShareGPT Format: { “conversations”: [{“from”: “system”, “value”: “...”}, ..] }
- ChatML Format:
    - <|im_start|>system
    - You are a helpful assistant.<|im_end|>
    - <|im_start|>user
    - What is the capital of France?<|im_end|>
    - <|im_start|>assistant
    - The capital of France is Paris.<|im_end|>

The Masked Loss

- During pre-training, you compute loss on every token.
- During SFT, you only compute loss on the response tokens.
  - The instruction tokens are there for context, but the model is not penalized for “predicting” them incorrectly.

→ You don’t want the model to learn to generate instructions. You want it to learn to respond to instructions.

Training Hyperparameters

SFT uses dramatically different hyperparameters that pre-training.

Parameter	Pretraining (LLAMA 2 7B)	SFT (LLAMA 2 CHAT)
Learning rate	3e-4 (peak)	2e-5
Epochs	1 (single pass over data)	2
Batch size	4M tokens	64 examples
Warmup steps	2000	0 - 100
Weight decay	0.1	0.0-0.1
Data size	2T tokens	27000 examples

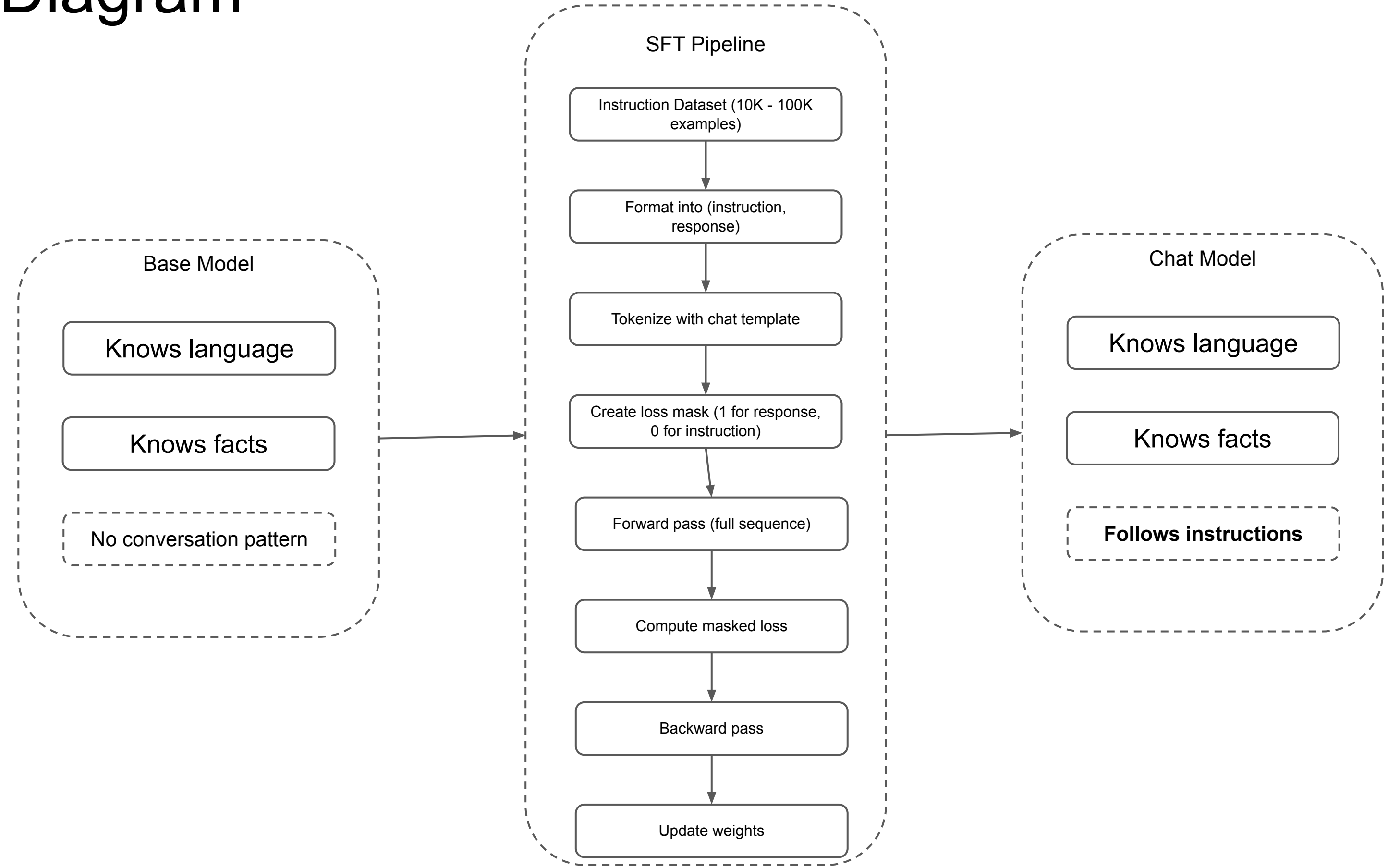
Catastrophic Forgetting

Fine-tuning can destroy general capabilities. Train too long on instruction-following data and the model losses its ability to do general tasks.

Three mitigations:

- Low learning rate: 1e-5 to 5e-5. Smaller updates mean less destruction of pre-trained features.
- Short training: 1-3 epochs. Stop before the model overfits.
- Mix in pre-training data: Llama 2 Chat mixed a small percentage (2-5%) of raw pre-training data into the SFT dataset. This “reminds” the model of its general capabilities while learning the new behavior.

Diagram



Base Model
Knows language
Knows facts
No conversation pattern
SFT Pipeline
Instruction Dataset (10K - 100K examples)
Format into (instruction, response)
Tokenize with chat template
Create loss mask (1 for response, 0 for instruction)
Forward pass (full sequence)
Compute masked loss
Backward pass
Update weights
Chat Model
Knows language
Knows facts
Follows instructions